# PART -1

# (2) Independent Project Report

**Sales Analysis and Profit Prediction using Machine Learning**

**1. Introduction**

This project focuses on analysing sales data from a retail store using machine learning techniques. The dataset contains information about customer orders, product categories, sales, profit, and regional distribution. The goal of the project is to perform exploratory data analysis (EDA), preprocess the data, train a machine learning model, and evaluate its performance in predicting profit.

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**2. Dataset Description**

The dataset used for this project is the Superstore Sales Dataset obtained from Kaggle. It contains information related to sales transactions of a retail store.

Key characteristics of the dataset:

- Number of rows: 9994
- Number of columns: 21
- Data types: Integer, Float, and Object (categorical)

Important features in the dataset include:

- Order ID
- Customer Name
- Category and Sub-Category
- Sales
- Quantity
- Discount
- Profit
- Region and State

The dataset did not contain any missing values, making it suitable for analysis and modeling.

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**3. Exploratory Data Analysis (EDA)**

Exploratory Data Analysis was performed to understand patterns and relationships in the dataset.

Some visualizations created include:

**Sales by Category**
This graph shows that the Technology category generates the highest sales, followed by Furniture and Office Supplies.

**Sales by Region**
The analysis shows that sales are relatively balanced across regions, with the South and East regions having slightly higher sales.

**Profit Distribution**
The histogram of profit values shows that most transactions have small profit values, while a few transactions show large profits or losses.

**Sales vs Profit Relationship**
A scatter plot was used to analyse the relationship between sales and profit.

These visualizations help in understanding the sales trends and customer purchasing patterns.

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**4. Data Preprocessing**

Before training the machine learning model, data preprocessing was performed.

Steps included:

- Checking for missing values
- Selecting numerical features such as Sales, Quantity, Discount, and Profit
- Splitting the dataset into training and testing sets

The dataset was divided into:

- **80% training data**
- **20% testing data**

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**5. Model Training**

A scikit-learn Linear Regression model was used to predict profit based on input features such as sales, quantity, and discount.

Linear Regression is a supervised machine learning algorithm used to model the relationship between dependent and independent variables.

The model was trained using the training dataset.

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### 6. Model Evaluation

After training the model, predictions were made on the testing dataset.

Two evaluation metrics were used:

**Mean Squared Error (MSE)**
Measures the average squared difference between actual and predicted values.

**$R^2$ Score (Coefficient of Determination)**
Indicates how well the model explains the variance in the data.

A higher $R^2$ value indicates better model performance.

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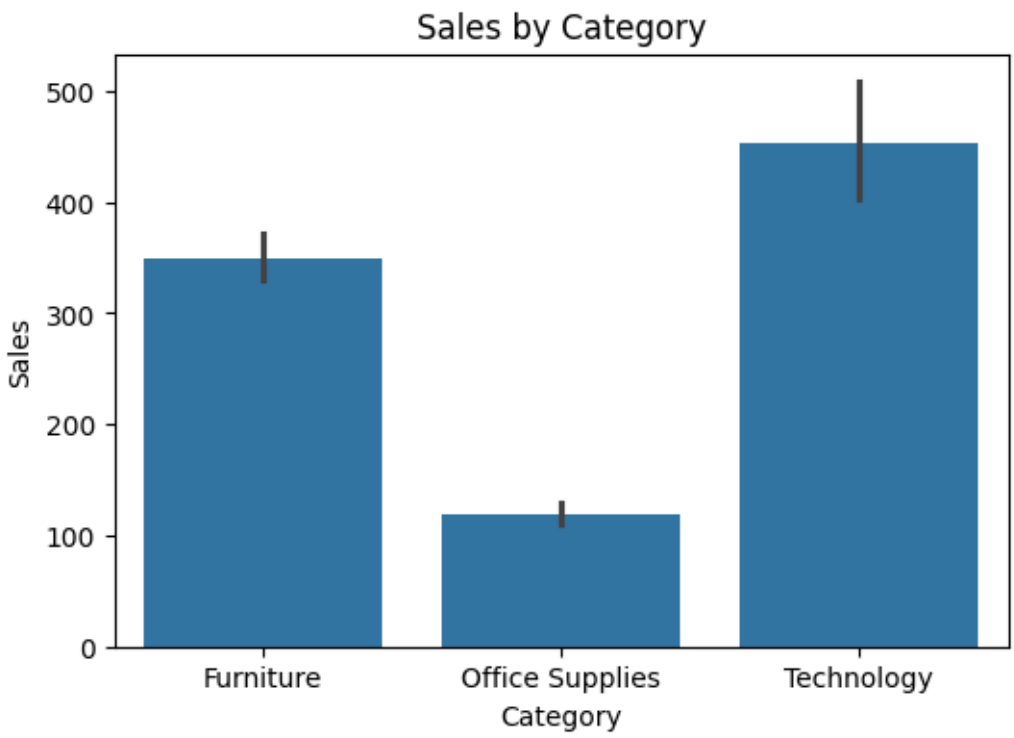
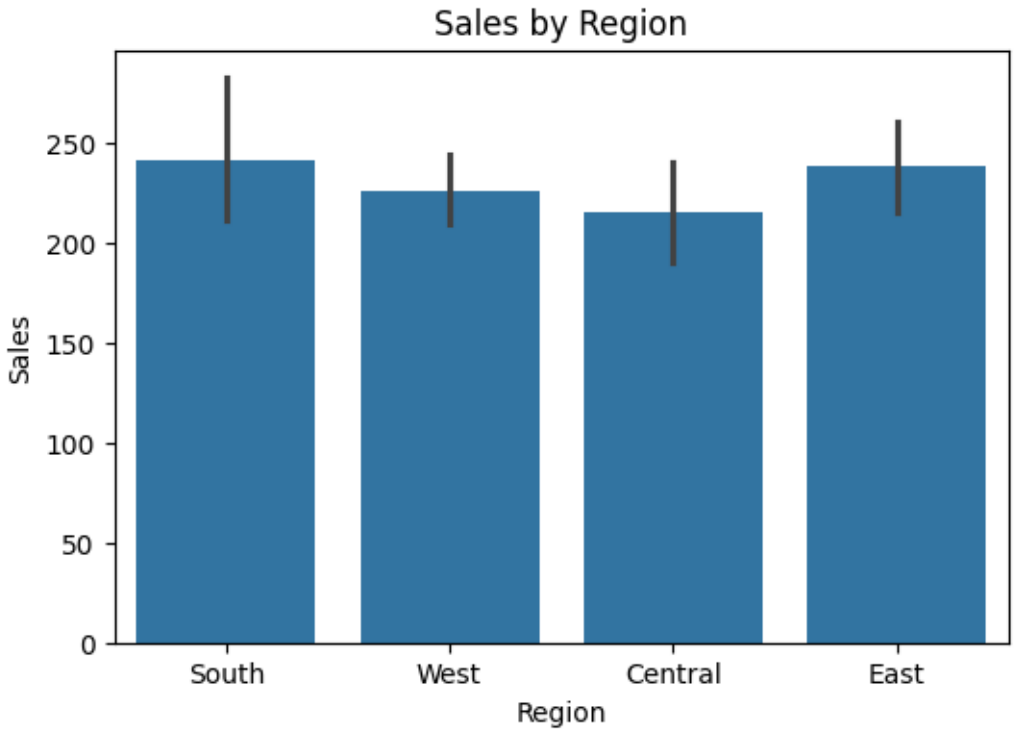
### 7. Results

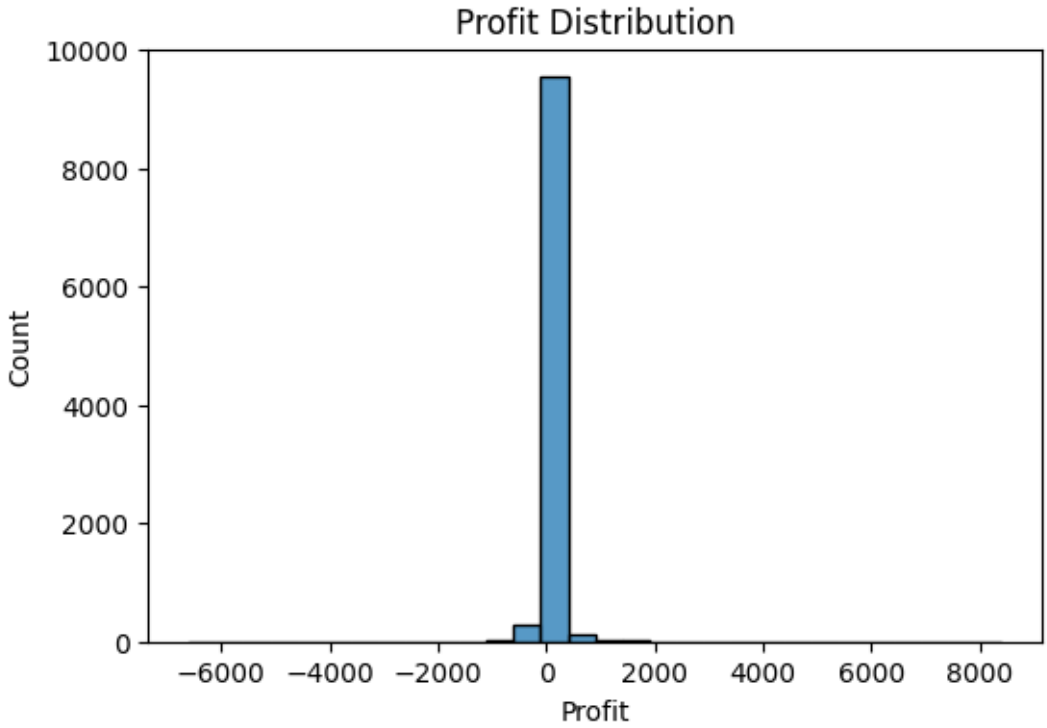
The model successfully predicted profit values based on the input features. The evaluation metrics show that the model is able to capture the relationship between sales-related variables and profit.

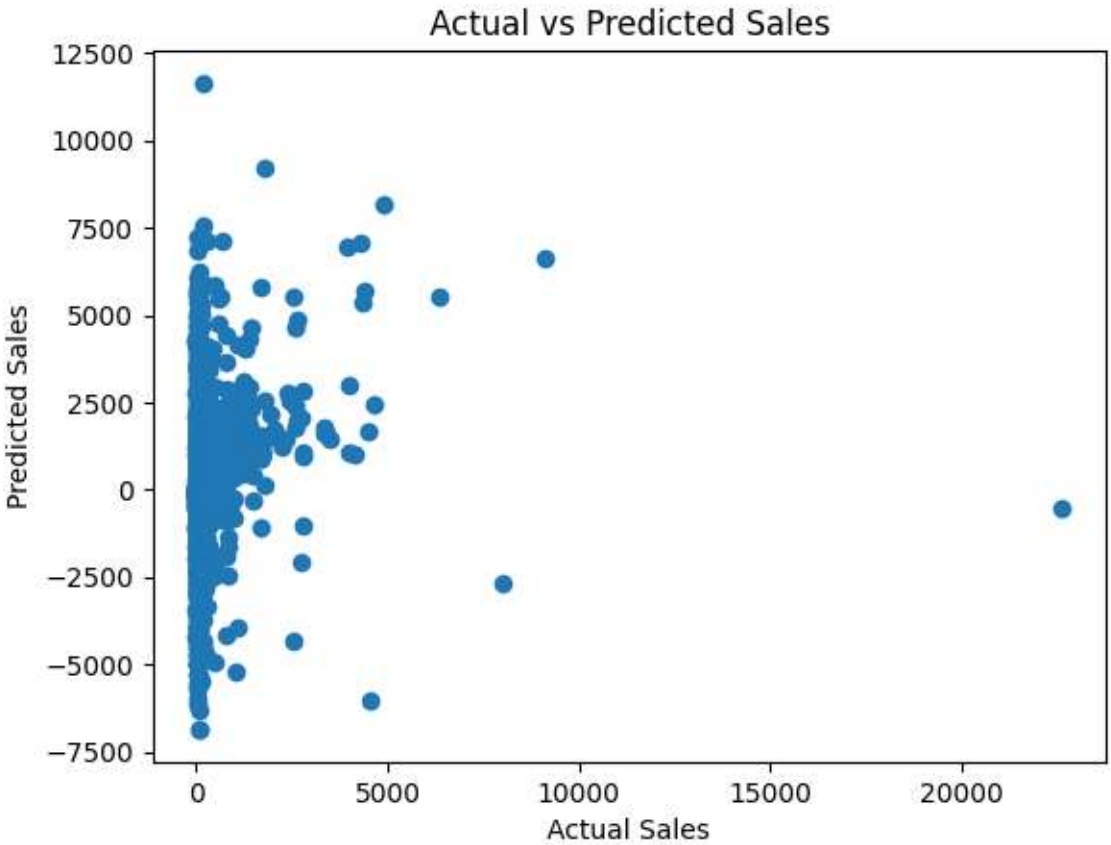
Although the model provides reasonable predictions, more complex models and additional features could improve prediction accuracy.

```
Sample - Superstore.csv(text/csv) - 2287806 bytes, last modified: 3/7/2026 - 100% don
Saving Sample - Superstore.csv to Sample - Superstore (19).csv
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```


Sales by Category
500
400
300
200
100
0
Sales
Furniture
Office Supplies
Technology
Category


Sales by Region
250
200
150
100
50
0
Sales
South
West
Central
East
Region


Profit Distribution
Count
Profit
10000
8000
6000
4000
2000
0
−6000
−4000
−2000
0
2000
4000
6000
8000


Actual vs Predicted Sales
Predicted Sales
Actual Sales
12500
10000
7500
5000
2500
0
−2500
−5000
−7500
0
5000
10000
15000
20000

## 8. Conclusion

In this project, sales data from a retail store was analyzed using exploratory data analysis and machine learning techniques. Several visualizations were created to understand sales patterns across categories and regions.

A Linear Regression model was trained to predict profit using features such as sales, quantity, and discount. The model evaluation showed that machine learning can be effectively used to analyse retail sales data and generate useful predictions.

Future improvements could include using more advanced machine learning models and incorporating additional features for better prediction accuracy.